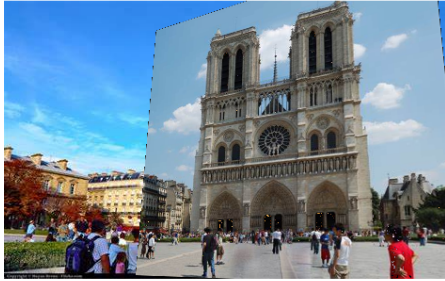


Crowdsourced Streetview: Integrating Real-Time Imagery Updates into Google Streetview

Ryan Hardesty Lewis
rl869@cornell.edu
Cornell Tech
New York, NY, USA



Notre Dame, France



Tower of Pisa, Italy



St. Mark's Basilica, Italy

Figure 1: Our system generates real-time Streetview imagery via crowdsourced overlays. These are examples of Crowdsourced Streetview integrating user-contributed images into existing Google Streetview data.

ABSTRACT

This paper presents Crowdsourced Streetview, a system that integrates real-time imagery updates into Google Streetview by leveraging crowdsourced images from social media platforms. Our approach utilizes advanced image alignment and feature detection algorithms to overlay user-contributed images onto existing Streetview data, achieving real-time performance with an image alignment runtime of 19ms per image pair. Crowdsourced Streetview has the potential to provide up-to-date and real-time visual representations of locations, particularly in rapidly changing environments and during events such as natural disasters.

ACM Reference Format:

Ryan Hardesty Lewis. 2024. Crowdsourced Streetview: Integrating Real-Time Imagery Updates into Google Streetview. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671067>

1 INTRODUCTION

Google Streetview provides users with panoramic views of streets worldwide but often contains outdated imagery that fails to reflect the current state of the environment. This limitation is particularly evident in rapidly changing urban landscapes or in the aftermath

of significant events. Crowdsourced Streetview addresses this issue by integrating real-time imagery updates from user-contributed photos on social media platforms, enhancing the accuracy and timeliness of the Streetview experience.

Previous research has explored the integration of crowdsourced data into digital mapping systems. Heinly et al. [Heinly et al. 2015] proposed constructing 3D city models using images from Yahoo, while Mooney et al. [Mooney et al. 2013] discussed the potential of volunteered geographic information in updating spatial datasets. However, these approaches have not focused on real-time integration with street-level imagery, which is our primary focus.

The main contributions of this paper are threefold. First, we propose a novel system architecture for integrating real-time crowdsourced imagery updates into Google Streetview, enabling the platform to provide users with up-to-date visual representations of their surroundings. Second, we develop optimized image alignment and feature detection algorithms that achieve real-time performance, ensuring the seamless integration of user-contributed images into the existing Streetview data. Finally, we demonstrate the system's potential to provide timely and accurate visual information for rapidly changing environments, such as urban landscapes undergoing development or areas affected by natural disasters, showcasing the practical applications and benefits of our approach.

2 OUR APPROACH

Crowdsourced Streetview's architecture consists of several key components, but primarily comprises of keypoint matching, as illustrated in Figure 2. The image collection module gathers geo-tagged images from social media platforms using relevant hashtags and EXIF data. This allows the system to identify user-contributed images that are likely to be relevant to specific locations within the Streetview dataset and match against nearby imagery.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Posters '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0516-8/24/07

<https://doi.org/10.1145/3641234.3671067>

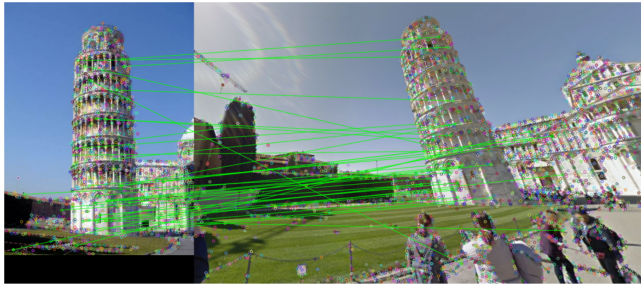


Figure 2: Matching features for proper placement and perspective warping using keypoints.

The image alignment component employs homography algorithms from the OpenCV library to accurately align the crowd-sourced images with the existing Streetview data. This process involves detecting and matching key features between the user-contributed image and the corresponding Streetview panorama. By leveraging optimized feature detection algorithms, such as SIFT and SuperPoint, and hardware acceleration through NVIDIA’s TensorRT library, the system achieves real-time performance with an image alignment runtime of just 19ms per image pair.

To be completely real-time, our system uses a controlled Puppeteer browser to intercept Google Streetview network calls and send the images to our Python server first, which processes image tiles into an equirectangular image, overlays any social media, and deconstructs back into tiles for the user’s browser. In doing this, we update Google Streetview’s interface with our own imagery.

Users can now explore the enhanced panoramas through the familiar Streetview interface. By continuously integrating new crowd-sourced images as they become available, Crowdsourced Streetview ensures that users have access to the most current and accurate visual information for any area.

3 EVALUATION AND DISCUSSION

To evaluate the performance of Crowdsourced Streetview, we conducted both quantitative and qualitative assessments of the system. Table 1 presents a comparison of the image alignment runtimes achieved by our approach using different feature detection algorithms. By leveraging optimized algorithms and hardware acceleration, Crowdsourced Streetview achieves a significant speedup, with TensorRT-accelerated SuperPoint feature detection enabling real-time performance, enabling us to update a user’s browser live with relevant crowd-sourced imagery. We see exponential time increase because of the number of comparisons against social media, making even small improvements in algorithm time incredibly important.

Table 1: Image Alignment Performance Comparison

Algorithm	Runtime (1 Image)	Runtime (250 Images)
TensorRT SuperPoint	19ms	4.75s
ORB	0.3176s	79.40s
SIFT	1.2893s	322.33s

Qualitative evaluations were performed by visually inspecting the updated Streetview panoramas generated by our system. Figure 1 presents examples of Crowdsourced Streetview integrating user-contributed images into existing Streetview data.



Figure 3: Crowdsourced imagery and Google Streetview imagery are combined to create Crowdsourced Streetview.

Our system demonstrates the feasibility and potential benefits of integrating real-time crowd-sourced imagery updates into Google Streetview. By leveraging the vast amount of user-generated visual data available on social media platforms, Crowdsourced Streetview can provide more current and accurate representations of rapidly changing environments, especially those affected by disaster, as we can note by a combination of imagery in Figure 3.

However, there are several limitations and challenges associated with the current implementation of Crowdsourced Streetview. The quality and relevance of the crowd-sourced images can vary significantly, potentially affecting the accuracy and coherence of the updated panoramas. Future work could explore image filtering and content moderation techniques to ensure only the most pertinent and reliable user-contributed data is integrated.

Additionally, the scalability of the proposed approach may be a concern when dealing with the massive amounts of imagery data available on social media platforms. Further research into distributed processing architectures and advanced data management strategies could help address this issue and ensure that Crowdsourced Streetview can effectively handle large-scale, real-time updates with thousands of photos, rather than a sample few.

Despite these challenges, the successful integration of real-time crowd-sourced imagery updates into Google Streetview has the potential to revolutionize the way we interact with and understand our environment through digital mapping technologies. By providing users with up-to-date, dynamic visual representations of the world around them, Crowdsourced Streetview can enable new applications and insights in areas such as urban planning, emergency response, and virtual tourism.

By continuing to refine and expand the capabilities of Crowdsourced Streetview, we aim to contribute to the advancement of digital mapping technologies and enhance our understanding of the dynamic world around us on a real-time basis.

ACKNOWLEDGMENTS

We use data from Google Maps. © 2024 Google LLC, under fair use.

REFERENCES

- Jared Heinly, Johannes L. Schönberger, Enrique Dunn, and Jan-Michael Frahm. 2015. Reconstructing the World in Six Days. In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 3287–3295. <https://doi.org/gngd2x>
- Peter Mooney, Padraig Corcoran, and Blazej Ciepluch. 2013. The potential for using volunteered geographic information in pervasive health computing applications. *Journal of Ambient Intelligence and Humanized Computing* 4, 6 (2013), 731–745. <https://doi.org/10.1007/s12652-012-0149-4>